Concepts

System bus: All the components and peripherals inside the computer such as the CPU, RAM, hard disk, graphics card, etc. communicate with each other via a data highway, generally referred to as the system bus or simply the bus. With continued improvements in technology driven by the need for faster communication between the various components, the bus has evolved over time. The older buses were ISA, EISA, MCA and VLB. We now live in the era of the PCI, AGP and PCI-Express (PCIe) buses. Each is faster than the previous one.

PCI: The most common bus on modern motherboards, with a maximum transfer rate of 133 MBps. The major drawback with PCI is that it is an 'open' highway. That is, all devices use the same highway mostly on a first-come-first-served basis. If the bus is fully in use, the devices would have to wait in queue till they get their chance. It's too slow for the graphics-rich 3D applications of today.

AGP: Strictly speaking, AGP is not a bus but a port. However, in terms of computing, its behaviour is more like a bus. AGP is a special kind of bus designed for exclusive use by graphics cards. This frees up the PCI bus to handle the data exchange between the various other components. AGP 8x has a maximum transfer rate of slightly more than 2 GBps.

PCIe: This is the new standard in graphics and system buses. The increasing high-bandwidth demands of modern applications and devices have led to the development of the PCI-Express bus. To squeeze faster performance out of graphics cards, PCIe 16x delivers data transfer rates at a whopping 8 GBps. Here we won't explain how, but if you are curious, just remember that PCIe does this by "serial communication at higher clock rates" as against the parallel communication used by all earlier buses.